



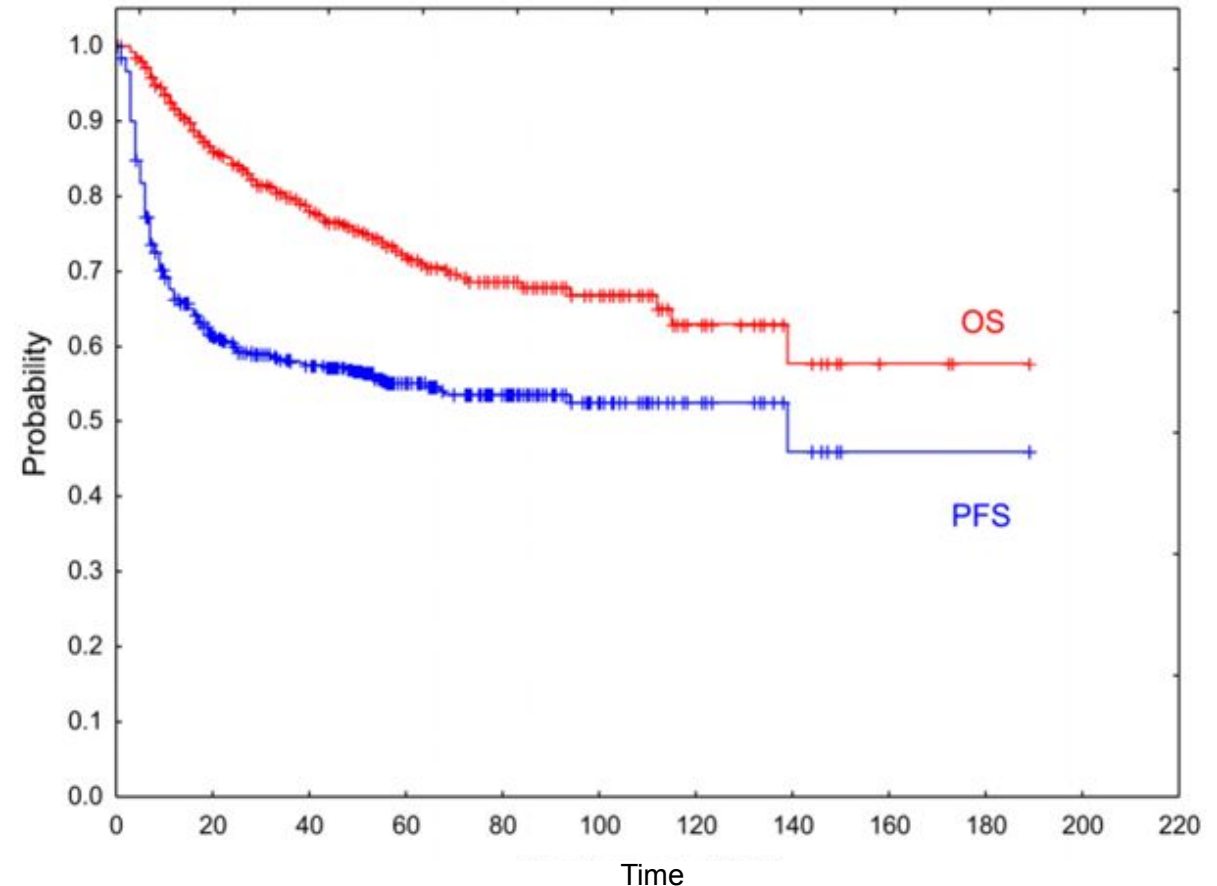
Developing decision models in R using published survival curves

Economic evaluation in oncology: why long-term outcomes matter

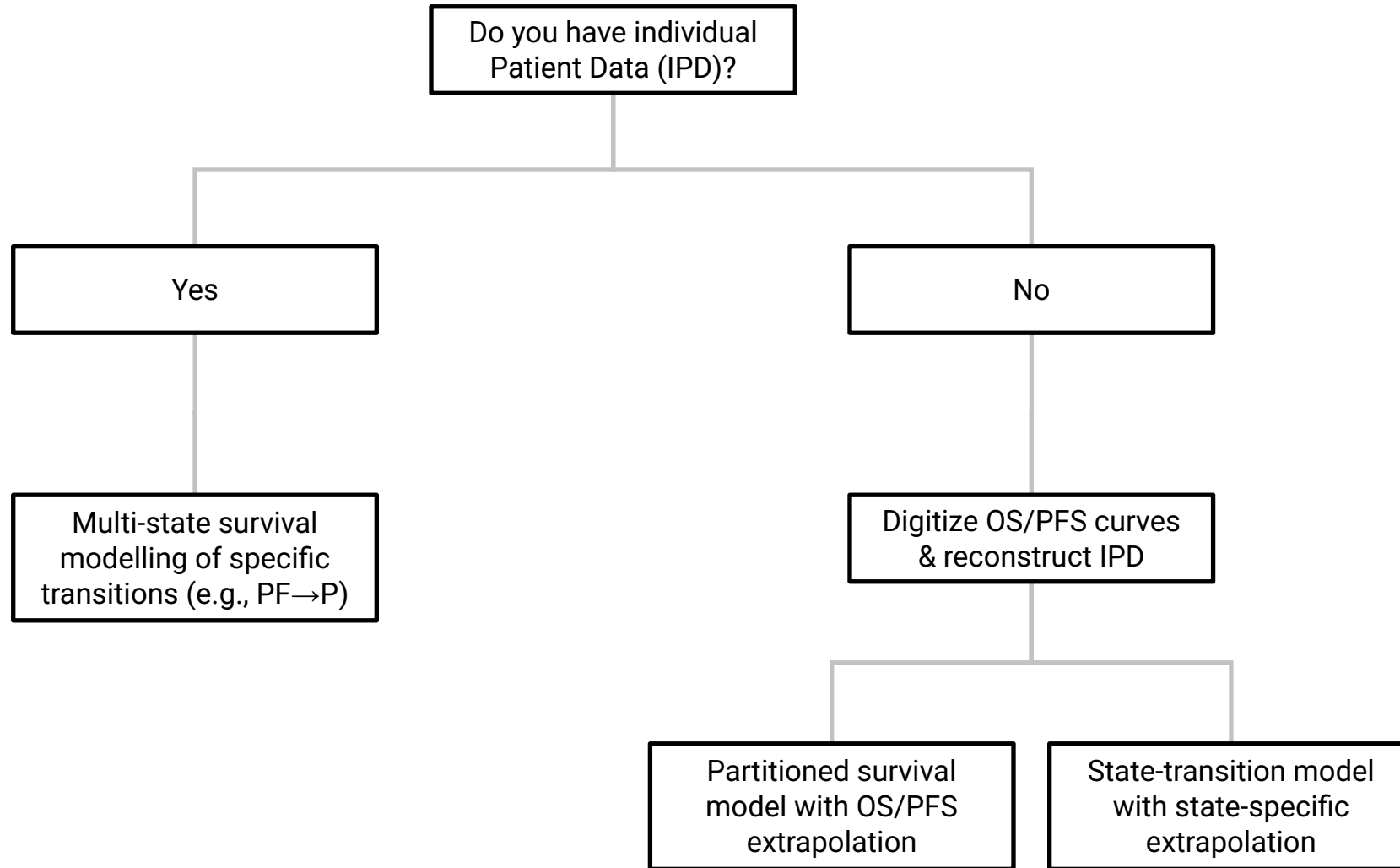
- Reimbursement decisions require evidence on long-term incremental costs and consequences of interventions
- Rapid growth of novel cancer therapies in the last two decades → expensive, evolving, and sometimes, impactful
- RCTs provide short-term follow-up
- Extrapolations (and corresponding estimates of uncertainty) are needed for decision making

Common endpoints in oncology

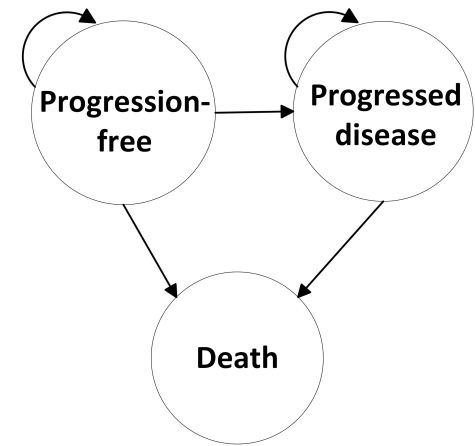
- **Overall Survival (OS):** time to death (any cause)
- **Progression-free survival (PFS):** time to disease progression or death



What is our approach?



If IPD is available:



Build a state-transition model (or discrete event simulation)

- State transition modeling: a stochastic process, $M(t)$, describing *transitions between states* over time
- Transitions occurring in discrete cycles (e.g., monthly) based on assumed **transition probabilities**
- Estimated and extrapolated through continuous *parametric multistate models*

If IPD is not available:

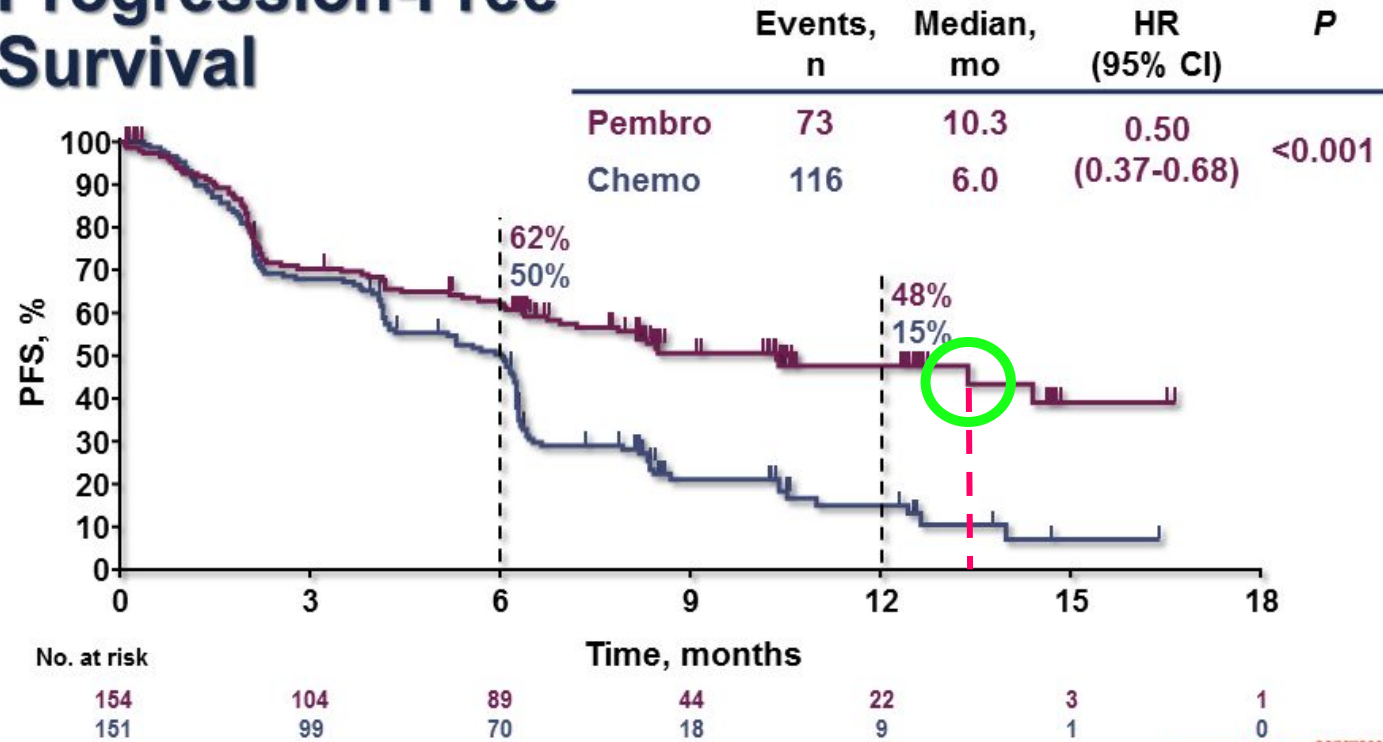
1. Digitize KM curves using point-and-click or automated software.
2. Reconstruct pseudo IPD based on algorithms.
3. Fit parametric models on pseudo IPD to regenerate OS/PFS.
4. Fit a *partitioned survival model* (PSM) (**or alternatives to PSM!**)

Methods for digitizing and reconstructing IPD

The concept of “Digitization”

To extract the time points where the survival probability changes and the magnitude of those changes

Progression-Free Survival



Assessed per RECIST v1.1 by blinded, independent central review.
Data cut-off: May 9, 2016.

MReck. ESMO 2016.

Available software

Point-and-click software

- Manual extraction
- Time intensive per-curve
- Potential for human error
- Reproducibility concerns

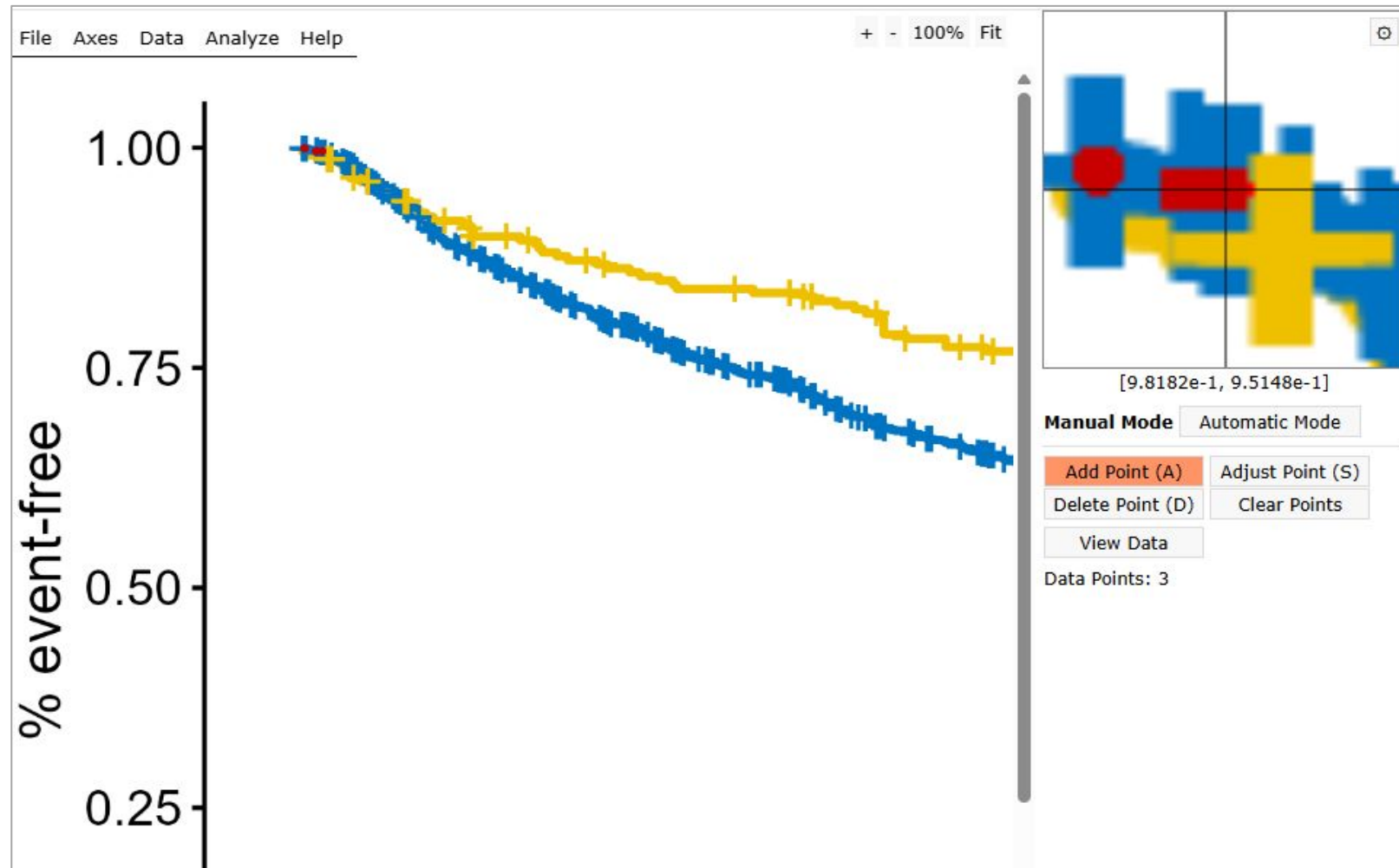
e.g. Digitizelt, WebPlotDigitizer/[Automeris.io](https://automeris.io),
extRpolateS

Automated software

- Minimal user input
- Increased accuracy
- May be behind paywall (\$)
- Integration within R

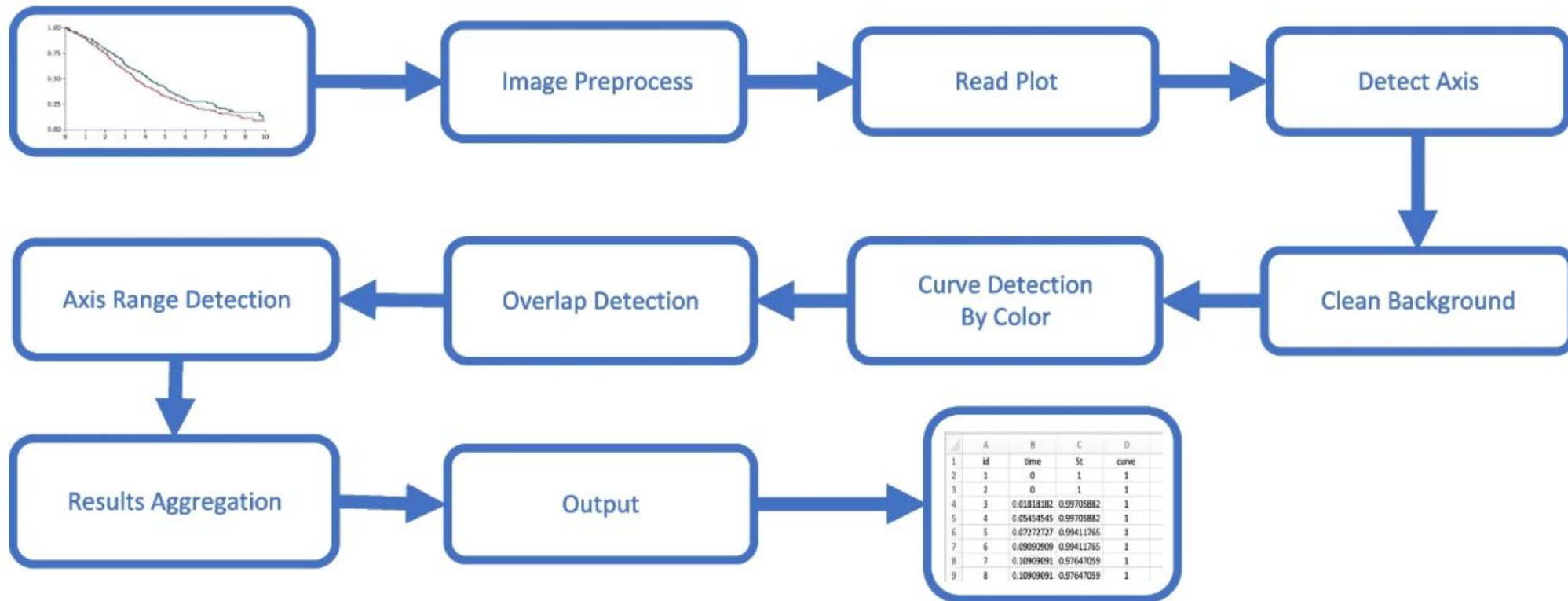
e.g. SurvDigitizeR, Engauge Digitizer, KM-GPT

Example of point and click software: *WebPlotDigitizer*



Automated software within R (and Shiny!)

SurvDigitizeR



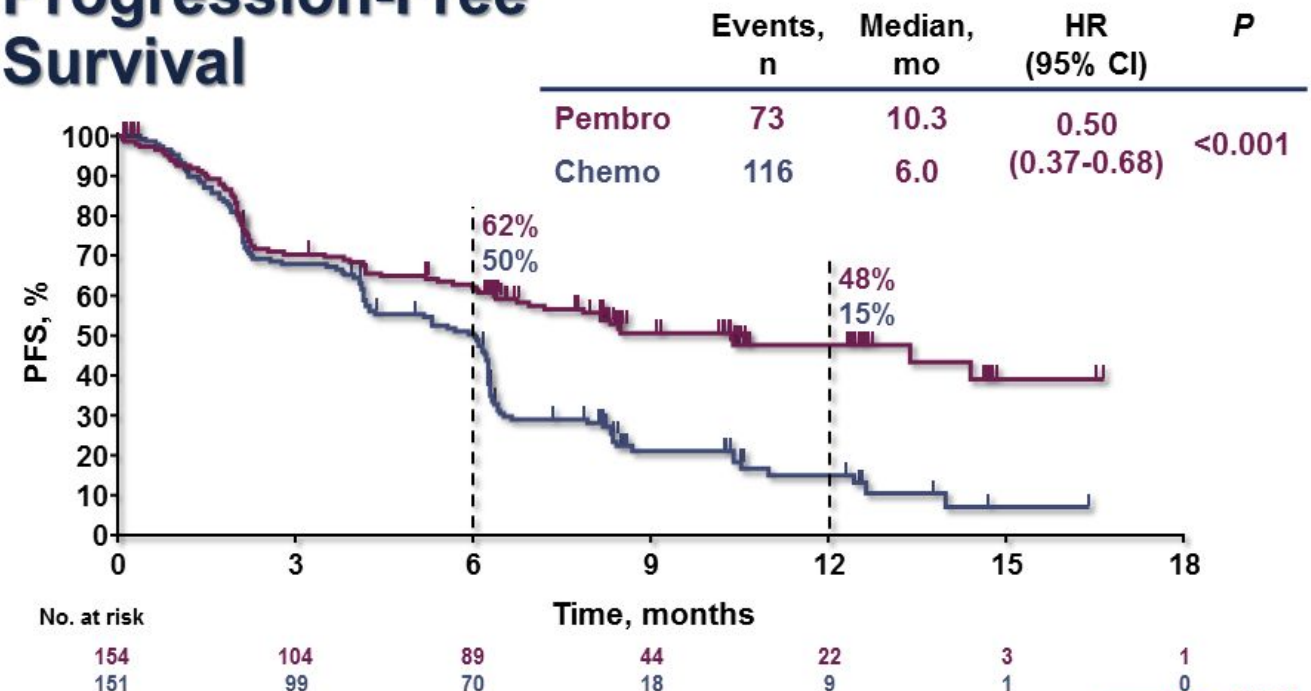
Algorithmic process for automated digitization

The concept of “Reconstructing” IPD

To “reverse-engineer” the underlying patient-level data from a published curve survival curve (the likely event and censoring times)

Create “pseudo IPD”

Progression-Free Survival



Assessed per RECIST v1.1 by blinded, independent central review.
Data cut-off: May 9, 2016.

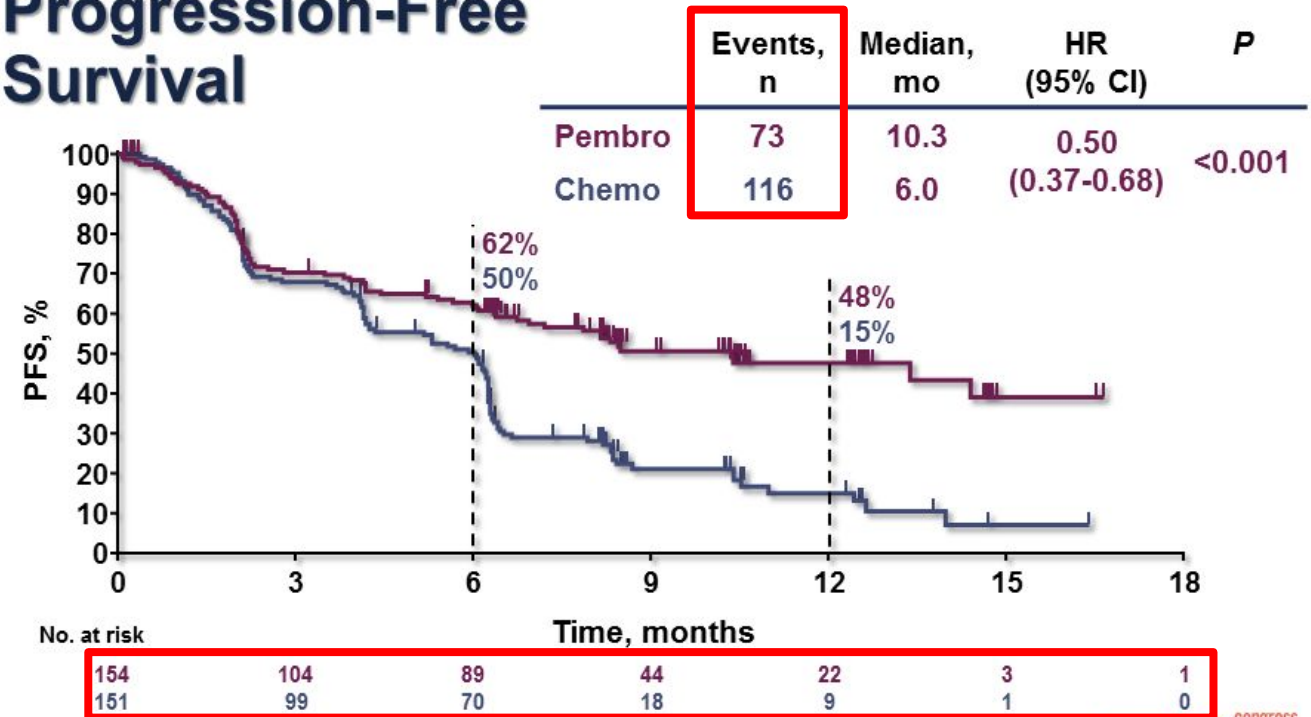
MReck. ESMO 2016.

The concept of “Reconstructing” IPD

Pertinent information:

- Survival probabilities at each “step” (via digitization)
- Total number of events and/or at-risk table

Progression-Free Survival



Assessed per RECIST v1.1 by blinded, independent central review.
Data cut-off: May 9, 2016.

Decision modeling
approaches utilizing
published survival curves

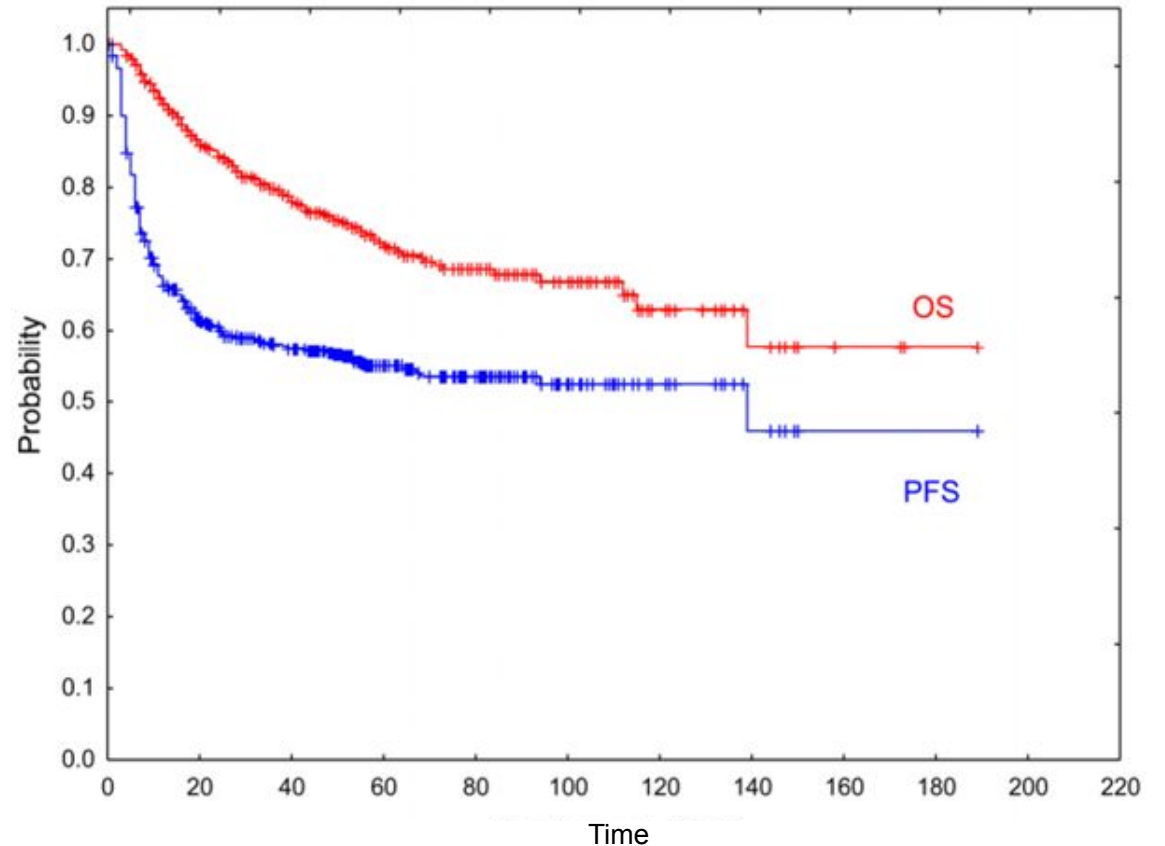
Partitioned Survival Modeling (PSM)

What is a partitioned survival model (PSM)?

A form of decision model that relies on evidence from **OS** and **PFS** curves

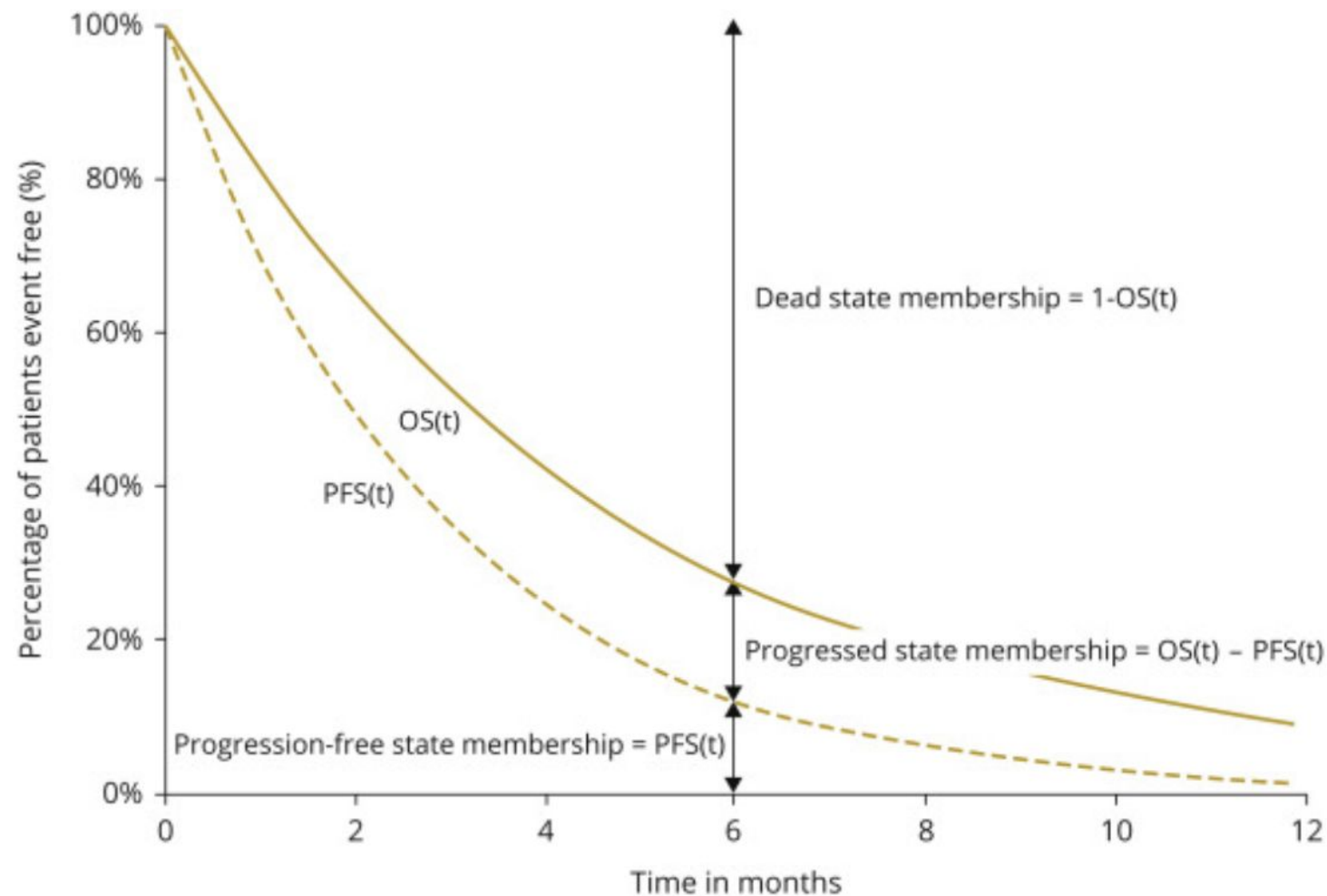
Allocation of a hypothetical cohort across distinct health states:

- Pre-progression
- Post-progression
- Dead



Partitioned survival models - mechanics

Information from OS & PFS curves is used to derive state membership

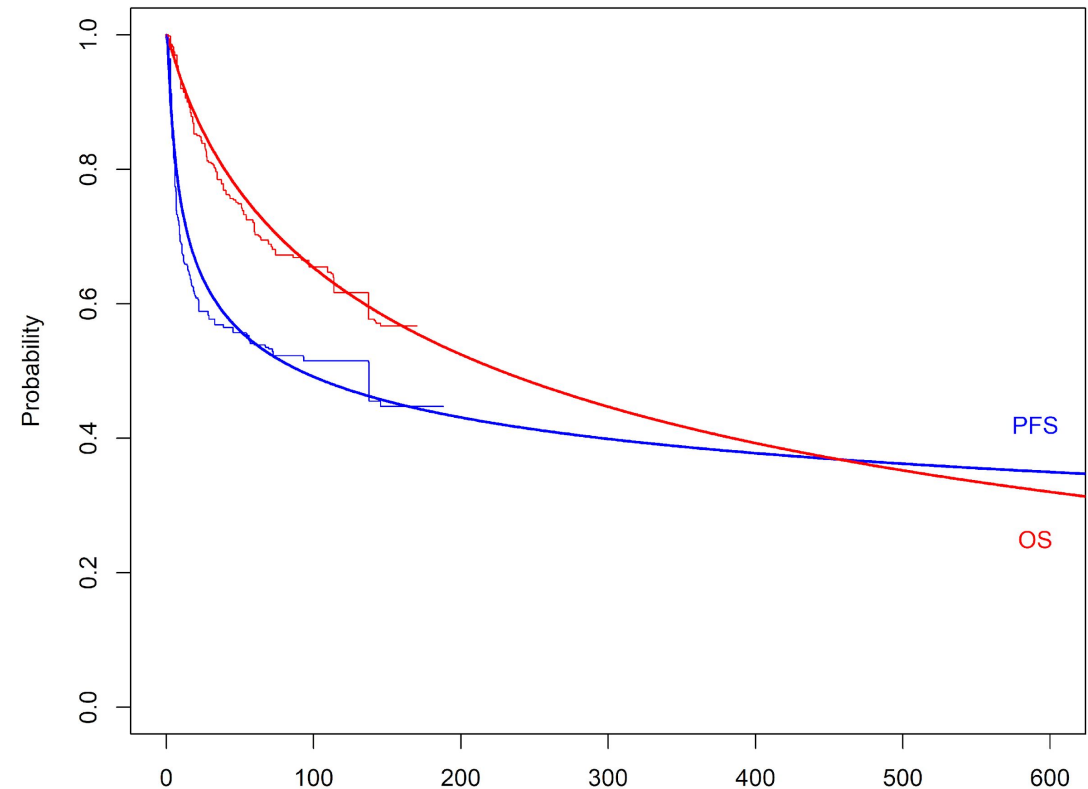


Partitioned survival models - the upsides

- Easy to communicate
- Easy to construct
- In-sync with what is commonly reported in RCTs
- In-sync with methods of cross-over
- Well-performing with sufficient follow-up (“mature” data)
- Can be more robust in data sources where certain transitions are rare

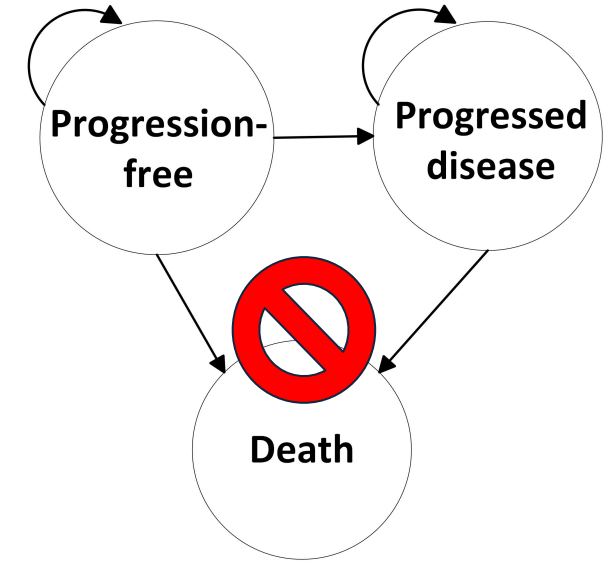
Partitioned survival models - the downsides

- Assumes PFS and OS are independent
- Cohort cannot transition back to a healthier state
- Difficult to accurately extrapolate in cases with few events and/or short observation period
- Potential for implausible extrapolations
- and more...

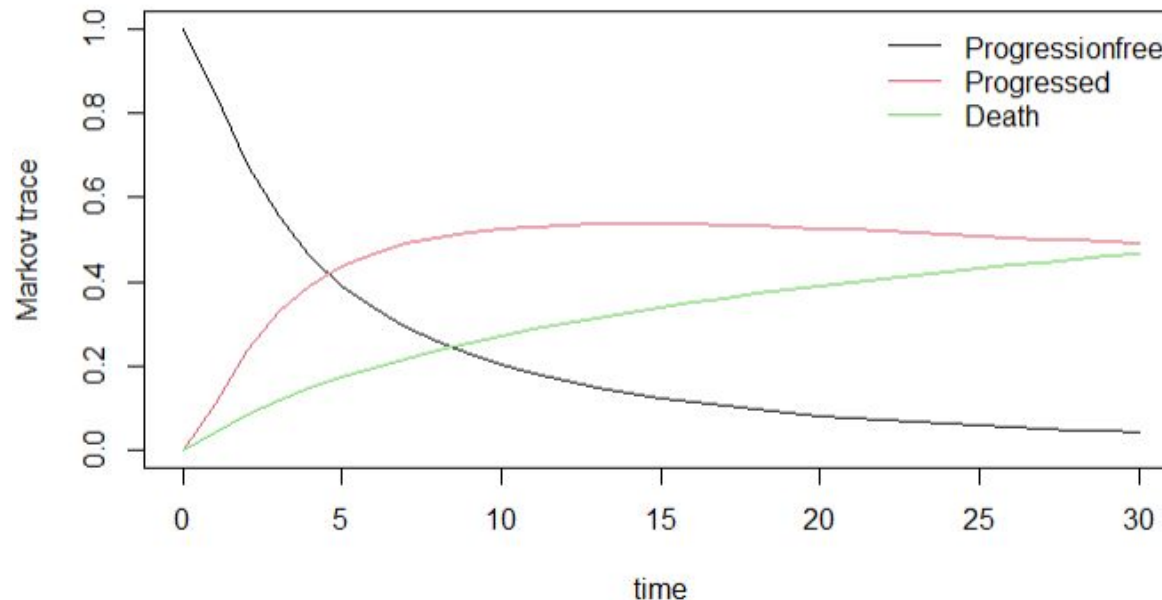


The main pitfall of PSM for decision modeling

Cannot distinguish how the cohort moves to the death state (PF→D vs. PD→D)



PFS: log-Normal, OS: log-Normal



```
> m_M_PSM[["trace"]]
      Progressionfree Progressed      Death
[1,]      1.000000000  0.0000000  0.0000000
[2,]      0.84458743  0.1116882  0.04372440
[3,]      0.67668424  0.2391910  0.08412475
[4,]      0.55320619  0.3286137  0.11818013
[5,]      0.46152256  0.3908559  0.14762151
[6,]      0.39154926  0.4348622  0.17358854
[7,]      0.33677553  0.4663871  0.19683738
[8,]      0.29297397  0.4891317  0.21789433
[9,]      0.25731540  0.5055431  0.23714149
[10,]     0.22784573  0.5172883  0.25486595
[11,]     0.20317645  0.5255340  0.27128950
[12,]     0.18229609  0.5311165  0.28658745
[13,]     0.16445176  0.5346472  0.30090101
[14,]     0.14907242  0.5365818  0.31434579
[15,]     0.13571776  0.5372645  0.32701774
```

Further reading on the PSM vs STM



Value in Health

Volume 23, Issue 12, December 2020, Pages 1613-1621



Methodology

Partitioned Survival and State Transition Models for Healthcare Decision Making in Oncology: Where Are We Now?

Beth S. Woods MSc¹  , Eleftherios Sideris MSc¹, Stephen Palmer MSc¹, Nick Latimer PhD², Marta Soares PhD¹

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² School of Health and Related Research, University of Sheffield, Sheffield, UK

NICE DSU TECHNICAL SUPPORT DOCUMENT 19: PARTITIONED SURVIVAL ANALYSIS FOR DECISION MODELLING IN HEALTH CARE: A CRITICAL REVIEW

REPORT BY THE DECISION SUPPORT UNIT

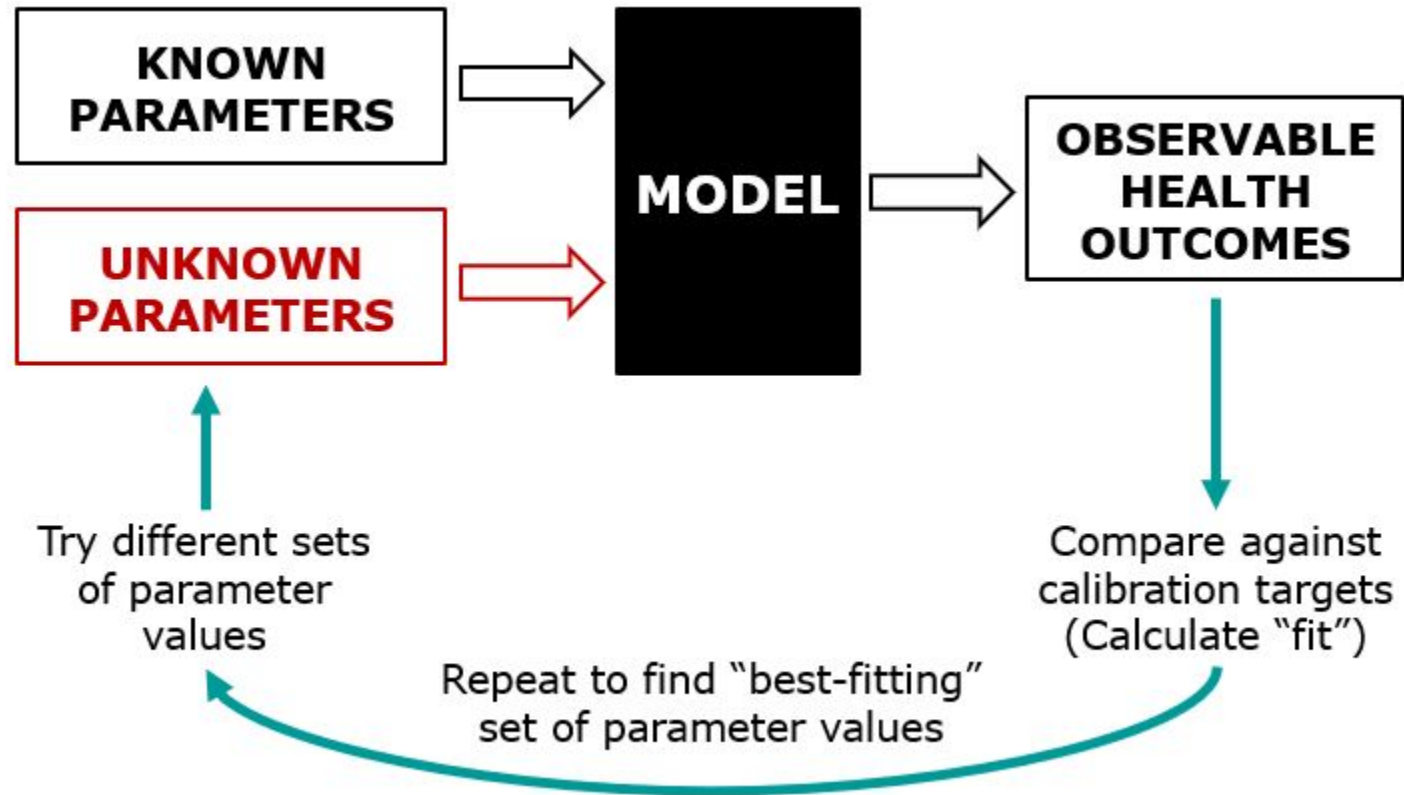
2 June 2017

Beth Woods¹, Eleftherios Sideris¹, Stephen Palmer¹, Nick Latimer², Marta Soares¹

Alternatives to PSM?

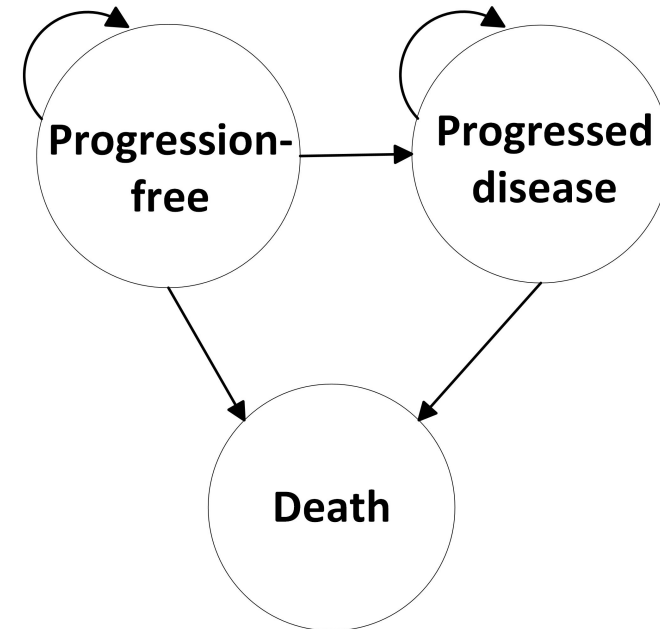
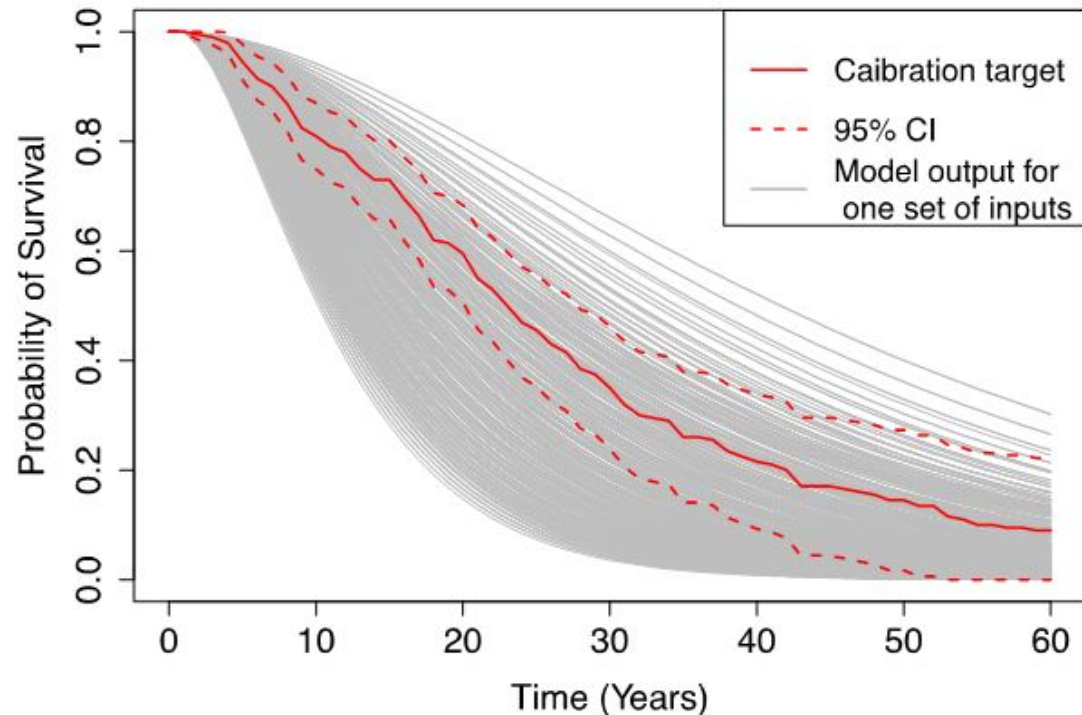
Methods to parameterize
state-transition models from
published survival data

The general process of calibration



What is calibration used for in this case?

Adjust model input parameters (i.e., transition rates) to match data on outcomes of interest (in this case, OS and PFS)



An analogy - figuring out a pancake recipe

Let's say that a friend makes a DELICIOUS pancake, but they don't want to share the recipe with you.



An analogy - figuring out a pancake recipe

You only know:

- how the pancakes look, smell, and taste at the end $[y]$, and
- the cooking process $[f()]$

How do you figure out how much of each ingredient to add (x) to get the same taste (y) ?



An analogy - figuring out a pancake recipe

First, we assume plausible measurements for the key ingredients

Each combination of ingredients is named as x_1



An analogy - figuring out a pancake recipe

Then we cook multiple combinations of ingredient measurements $[f(x_1), f(x_2), \dots, f(x_n)]$.

We will taste the results and identify which combinations generate the closest taste to what we are looking for (y).



Key concepts of Bayesian theorem

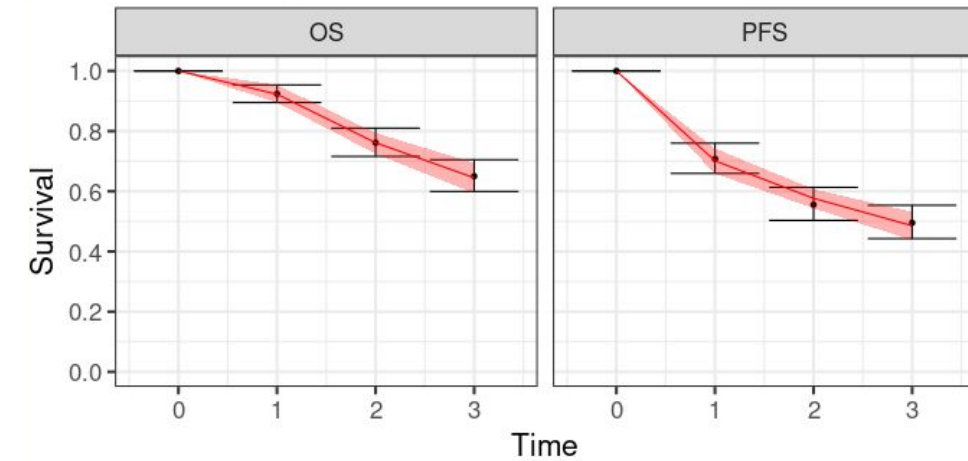
- ***Prior probability*** – The initial estimate of the probability of a hypothesis before observing the current evidence
- ***Likelihood*** – The probability that you would observe the specific data, assuming that your hypothesis is completely true
- ***Posterior probability*** – The updated probability, after taking the new data into account.

$$\text{Posterior probability} = \frac{\text{Likelihood} \cdot \text{Prior probability}}{\text{Normalization factor}}$$

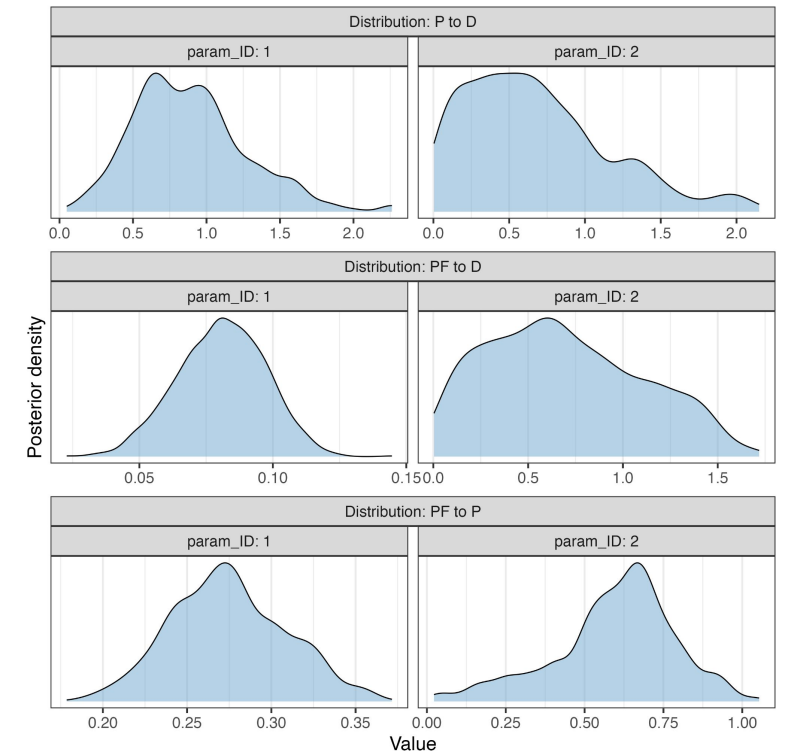
A bayesian calibration approach

DARTH workgroup

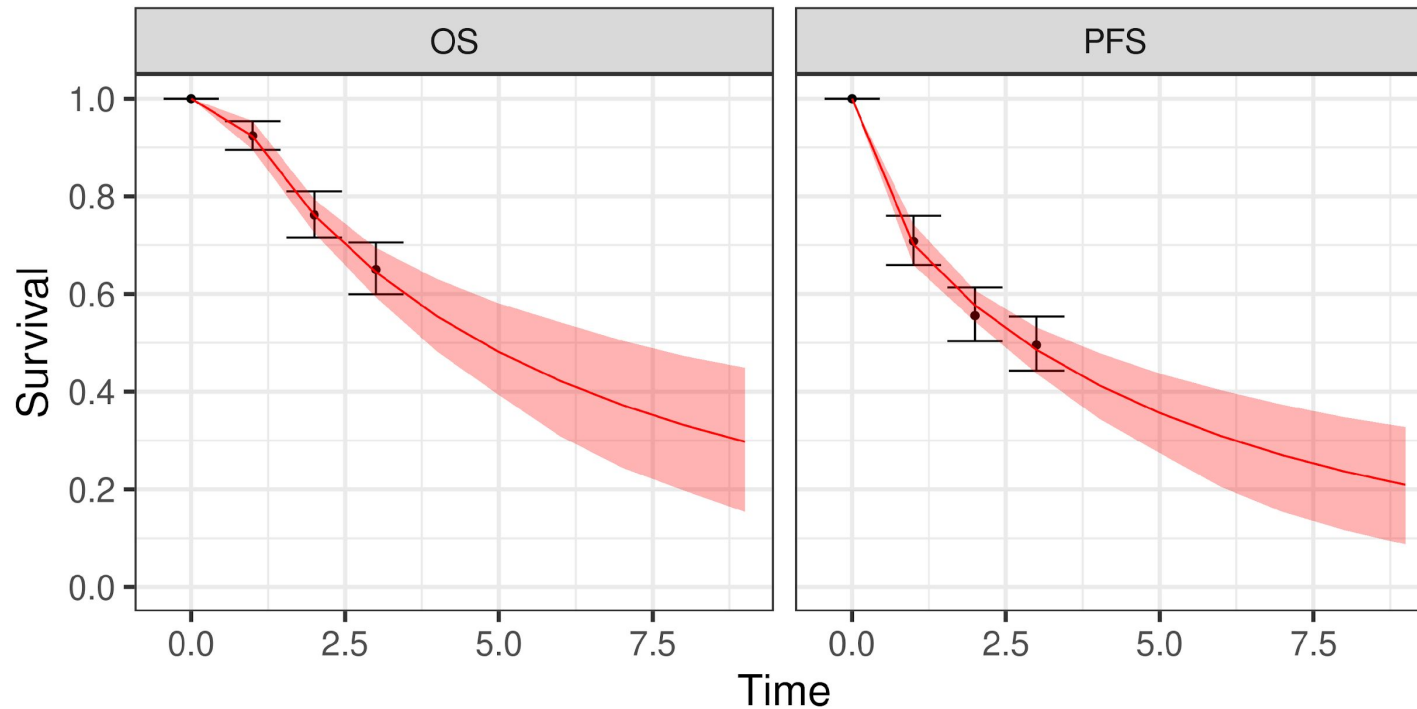
1. Digitize KM curves
2. Extract calibration targets (OS & PFS)
3. Assume parametric distributions for each transition hazard of a 3-state illness-death model
4. Assume priors for parameters of distributions
5. Model-predicted OS & PFS given parameter sets
6. Assess GoF & select posterior distributions
7. Assess assumptions and extrapolate model outcomes



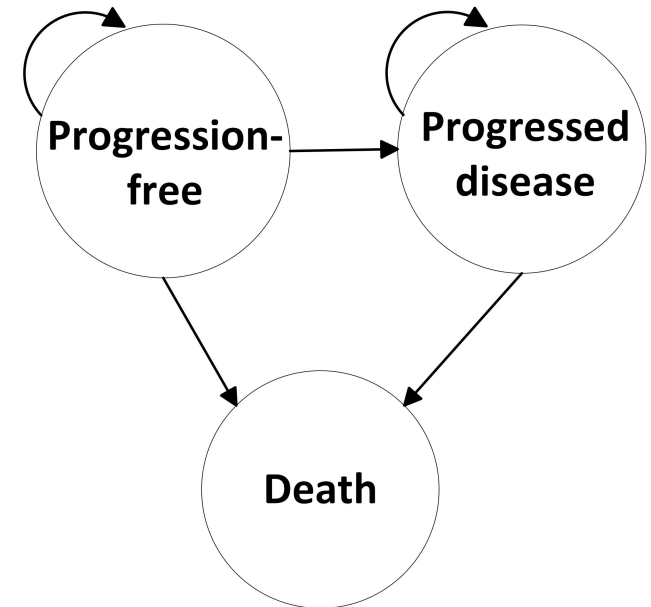
Source: ■ Model ● Targets



Extrapolate based on posteriors and model structure

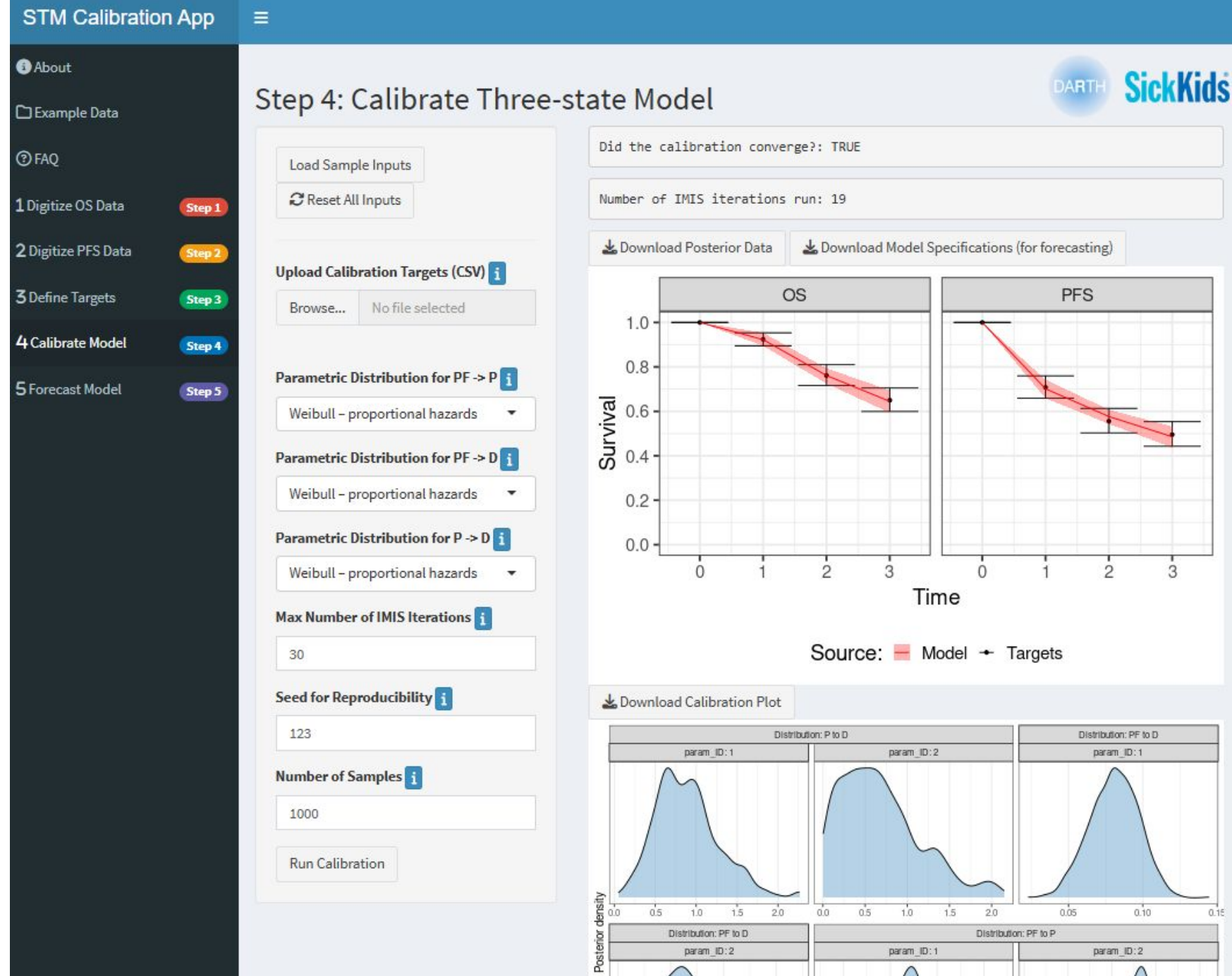


Source: — Model • Targets



Available as a Shiny app (beta version)

https://pechlilab.shinyapps.io/STM_calibration/



A case example



► Nat Commun. 2021 Dec 15;12:7304. doi: [10.1038/s41467-021-27358-6](https://doi.org/10.1038/s41467-021-27358-6)

Artificial intelligence-aided clinical annotation of a large multi-cancer genomic dataset

[Kenneth L Kehl](#)^{1,2,3,✉}, [Wenxin Xu](#)^{1,2,3}, [Alexander Gusev](#)^{1,2,3}, [Ziad Bakouny](#)^{1,2,3}, [Toni K Choueiri](#)^{1,2,3}, [Irbaz Bin Riaz](#)⁴, [Haitham Elmarakeby](#)^{1,2,3,5}, [Eliezer M Van Allen](#)^{1,2,3,5}, [Deborah Schrag](#)⁶

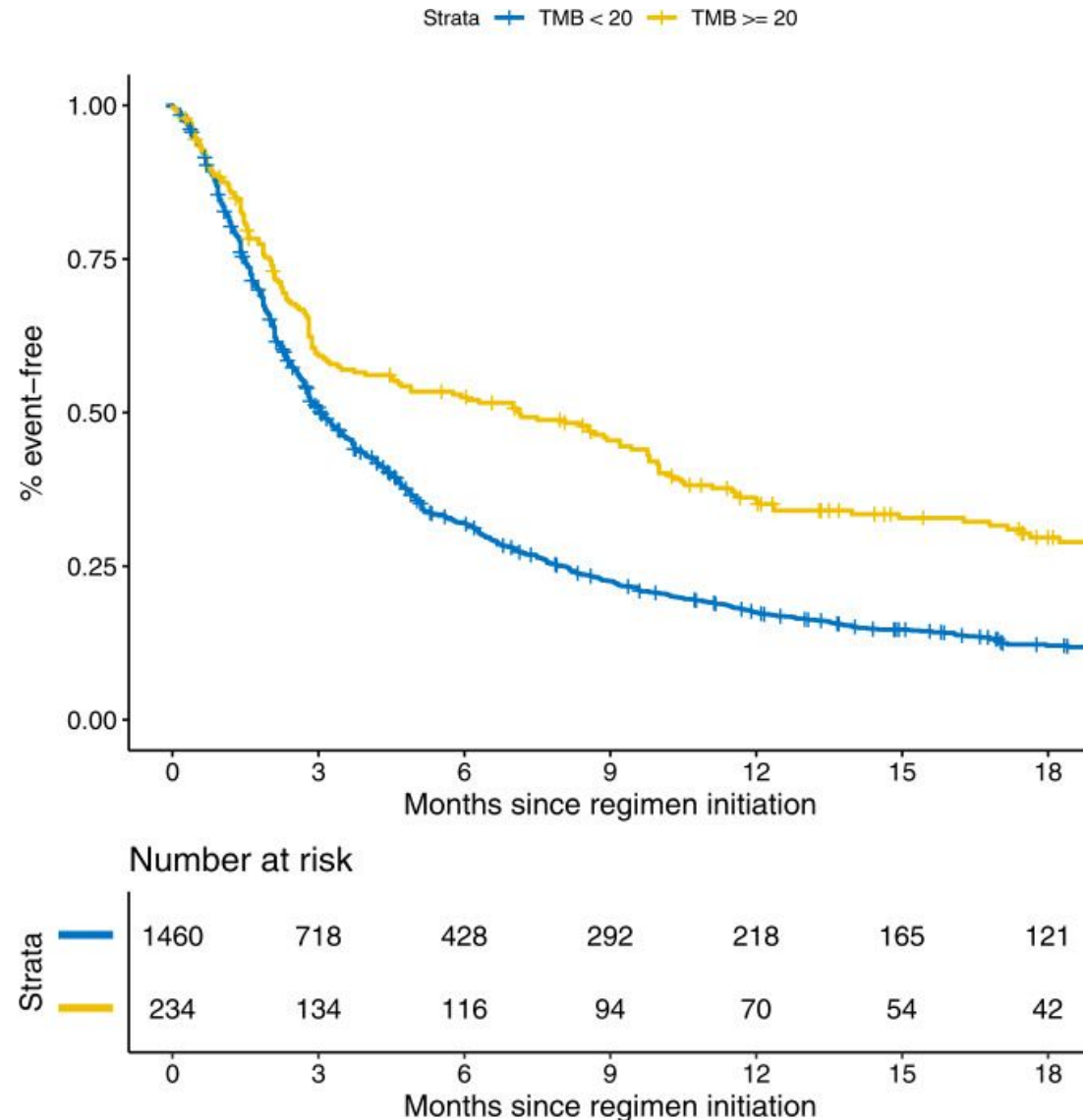
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PMCID: PMC8674229 PMID: [34911934](https://pubmed.ncbi.nlm.nih.gov/34911934/)

TMB has been characterized as a predictive biomarker for patients receiving immunotherapy.

What is the association between progression-free survival (PFS) and tumor mutation burden (TMB)?

OS and PFS curves were published



Patients on immunotherapy:

Group A: Those with high TMB
(≥ 20 mutations/megabase)

Group B: Those with low TMB
(< 20 mutations/megabase)

Advantages of a calibrated state-transition model

- Use of a 3-state model structure
 - more natural extrapolation process than composite OS/PFS
- Incorporation of uncertainty in state transitions
- Allow for intervention effects to be tested on specific transitions (e.g., apply a relative hazard on progression rate)